Dinosaur Poster or Slide Key

U02 / S02.3

COMMON PICKS · QUICK-CHECK FACTS

Fifteen dinosaurs students are likely to choose, from the 2026-10-27 research report. Hip: L = lizard-hipped (saurischian), B = bird-hipped (ornithischian). Lengths in meters. Fossil sites list solid finds only. For a dinosaur not in this table, check the student's facts against their source.

DINOSAUR	PERIOD	MYA	HIP	DIET	LEGS	LENGTH	FOSSIL LOCATIONS
TRIASSIC · 252 TO 201 MYA · ALL LIZARD-HIPPED							
Herrerasaurus	Late Triassic	~231	L	Meat	2	3–6	Argentina
Eoraptor	Late Triassic	~231–228	L	Small animals, maybe plants (sources conflict)	2	~1	Argentina
Plateosaurus	Late Triassic	~214–204	L	Plants	2	~5–10	Germany; Switzerland; France
Liliensternus	Late Triassic	poorly dated; 228–202 across sources	L	Meat	2	~5	Germany
Coelophysis	Late Triassic (latest)	~206–201	L	Meat	2	2–3	USA: New Mexico
JURASSIC · 201 TO 145 MYA							
Dilophosaurus	Early Jurassic	~196–183	L	Meat	2	~6–7	USA: Arizona
Allosaurus	Late Jurassic	~155–147	L	Meat	2	~8.5 (largest ~9.7)	USA: Colorado, Utah, Wyoming; Portugal
Stegosaurus	Late Jurassic	~155–150	B	Plants	4	~7–9	USA: Colorado, Utah, Wyoming; Portugal
Diplodocus	Late Jurassic	~155–147	L	Plants	4	~24–26	USA: Colorado, Utah, Wyoming, Montana
Brachiosaurus	Late Jurassic	~154–153	L	Plants	4	~18–22	USA: Colorado, Utah, Wyoming, Oklahoma. No Africa dot.
CRETACEOUS · 145 TO 66 MYA							
Spinosaurus	Cenomanian (early Late Cretaceous)	~100–94	L	Mostly fish	2	~12–15	Egypt; Morocco (required dots). Niger also accepted (*S. mirabilis*, 2026, PROVISIONAL). Algeria optional, marked "?"
Velociraptor	Late Cretaceous (Campanian)	~75–71 (tentative)	L	Meat	2	~1.5–2	Mongolia; China (Inner Mongolia)
Tyrannosaurus rex	Late Cretaceous (Maastrichtian)	~68–66	L	Meat	2	~12–13	USA: Montana, South Dakota, Wyoming; Canada: Saskatchewan
Triceratops	Late Cretaceous (Maastrichtian)	~68–66	B	Plants	4	~7–9	USA: Montana, North Dakota, South Dakota, Wyoming; Canada: Alberta, Saskatchewan
Ankylosaurus	Late Cretaceous (Maastrichtian)	~68–66	B	Plants	4	~6–8	USA: Montana, Wyoming; Canada: Alberta

GRADING NOTES

- **Google Slide.** Same rubric and points as the poster. Copy-paste shows up as pasted text of any length, a pasted photo used as the drawing, a default theme or white template left untouched, and images with no source on the slide. Pasted text, an untouched theme and missing sources cost points on Clear, visual, color. A pasted photo with no drawing of their own is a 0 on Drawing.

- **Period dates.** The handout uses rounded dates: 252–201, 201–145, 145–66 mya. The ICS values are 251.902, 201.4, 145.0 and 66.0 Ma. The researcher could not re-read the ICS chart this session, so those are **PROVISIONAL** until checked. Rounded dates are full credit either way.
- **Triassic hip types.** Every Triassic common pick is lizard-hipped. No securely bird-hipped Triassic dinosaur is known. The best candidate, *Eocursor*, may really be Early Jurassic. Worth saying out loud: early dinosaurs were mostly lizard-hipped.
- **Hip type from a picture.** *Velociraptor* and birds are lizard-hipped, but their pubis points backward. A student who reads hip type off a *Velociraptor* skeleton will get it wrong. The hip-type row on the handout says "in most of them" for this reason.
- **Habitat and special feature** aren't in this table and were not researched for this key. Accept any reasonable answer from the habitat list that the student can point to a source for. **PROVISIONAL**: there is no checked habitat for 14 of the 15 common picks. *Spinosaurus*: river floodplain (river and delta systems).
- ***Coelophysis* dates.** A 2024 study puts it at ~206–201 (Rhaetian). Older books give ~225. Accept either.
- ***Eoraptor*.** Sources disagree on diet (Sereno lab: meat eater) and on whether it is an early theropod or an early sauropodomorph. It is lizard-hipped either way. Accept carnivore or omnivore.
- ***Liliensternus*** is the weakest of the common picks: its dates are poorly pinned. Accept any Late Triassic date.
- **Walks on.** 4 legs for *Stegosaurus*, the sauropods, *Triceratops* and *Ankylosaurus* is standard textbook knowledge. It was not separately re-sourced.
- **Map dots not verified.** *T. rex* in North Dakota and Alberta, and *Triceratops* in Colorado, are likely but were not confirmed. A dot there is **PROVISIONAL**. Decide before grading whether it costs points.
- **Map, both formats.** Students may draw or trace the map, glue on a printed one (poster), or paste one in (slide). The fossil dots and the key must be the student's own. The map row grades "not today's map" and dots in the right places, not accurate coastlines. Open question for a class reference map (report §4): Blakey's 90 Ma Commons map (CC BY-SA 3.0, low-res only) or a trace from Scotese & Wright 2018 PALEOMAP.
- **School bus.** A full-size Type C bus is 12.0 m (39.3 ft). Re-solved: *T. rex* 12–13 m is about 1 bus; *Brachiosaurus* 18–22 m is 1.5 to 1.8 buses; *Diplodocus* 24–26 m is 2 to 2.2 buses; *Velociraptor* 1.5–2 m is an eighth to a sixth of a bus.

SPINOSAURUS NOTES

- **Swim or wade? Unresolved.** Everyone agrees it lived around water and ate fish. Ibrahim et al. 2014 (Science) and 2020 (Nature, the fin-like tail) and Fabbri et al. 2022 (dense, diver-like bones) argue it hunted underwater. Henderson 2018, Hone & Holtz 2021 (a wading "shoreline generalist", like a giant heron), Sereno et al. 2022 ("not an aquatic dinosaur") and Myhrvold et al. 2024 (Fabbri's statistics unreliable; Fabbri's team says its result holds) argue it waded and ambushed from the shore. Student-safe line: "Scientists are still arguing about whether it swam underwater to hunt or waded in the shallows like a giant heron."
- **Holotype destroyed.** BSP 1912 VIII 19, found in Egypt's Bahariya Formation in 1912 and named by Stromer in 1915, was destroyed on the night of 24/25 April 1944 when a British bombing raid hit the Munich museum. Only Stromer's descriptions, drawings and a photograph survive. The Moroccan neotype (FSAC-KK 11888, Ibrahim et al. 2014) has a disputed status.
- **Fossil dots.** Egypt and Morocco are the required dots. A Niger dot is also accepted: a second species, *S. mirabilis*, was named in February 2026 from Niger's Farak Formation (~95 Ma; Sereno et al., Science 391). **PROVISIONAL**: one paper, not yet independently tested, and not decided for student pages.
- **Age.** The rocks are Cenomanian, the first stage of the Late Cretaceous, so the table says Late. Papers also use the informal label "mid-Cretaceous". Accept a student who writes "middle of the Cretaceous".
- **Posture.** The table says 2 legs. The 4-leg reconstruction (Ibrahim 2014) is older and now mostly rejected.
- **Size.** Length estimates run 12.6 to 18 m; this key uses 12 to 15. Mass estimates run from about 6 to 21 t, because no complete skeleton exists. Say "one of the longest meat-eating dinosaurs known." Do not let "the biggest meat-eater ever" onto a poster or slide as fact.

COMMON MISTAKES

STUDENTS SAY OR DRAW	CORRECTION
"Pterosaurs are flying dinosaurs."	Not dinosaurs. They are the dinosaurs' closest relatives.
"Plesiosaurs, ichthyosaurs and mosasaurs are swimming dinosaurs."	Marine reptiles, not dinosaurs. Mosasaurs are lizards (squamates).

STUDENTS SAY OR DRAW	CORRECTION
"*Dimetrodon* is a dinosaur."	A Permian synapsid, closer to mammals. It died out roughly 60 million years before the first dinosaurs.
Grassy plains behind the dinosaur.	Grass was absent or rare for most of the Mesozoic. Grasslands spread in the Cenozoic. Use ferns, conifers, cycads, and flowering plants in the Cretaceous.
T. rex fighting *Stegosaurus*.	At least 77 million years apart (~145–150 vs ~68 mya). *T. rex* lived closer in time to us than to *Stegosaurus*.
Spinosaurus fighting *T. rex*.	At least ~26 million years apart (93.9 – 68), and on different continents.
Movie-sized *Velociraptor* with scales.	Turkey-sized (~1.5–2 m, mostly tail) and feathered: quill knobs on its forearm (Turner et al. 2007). Movie raptors are closer to *Deinonychus*.
Dilophosaurus with a frill that spits venom.	No evidence for either. The real animal was bigger than the movie one, ~6–7 m.
"Bird-hipped dinosaurs became birds."	No. Birds came from lizard-hipped theropods.
Mussaurus as a Triassic dinosaur.	Now dated to the Early Jurassic, ~193 mya. Older books say Triassic.
The Berlin "*Brachiosaurus*" skeleton (Tanzania).	*Giraffatitan* since Taylor 2009. Real *Brachiosaurus* is North American. No Africa dot.
Dinosaurs and people at the same time.	Non-bird dinosaurs died out 66 mya. *Homo sapiens* is about 300,000 years old.
Hip type judged from a *Velociraptor* or bird skeleton.	Pubis points backward even though it is lizard-hipped.
One "official" length or weight for *Spinosaurus*.	Estimates run 12–18 m and 6–21 t. Teach the range.
"All dinosaurs are extinct."	Birds are living dinosaurs. All dinosaurs except birds died out 66 mya.